A **quiver** is an oriented graph $Q$ (which we will assume to be finite). A **representation** of $Q$ over a field $k$ is an assignment of a $k$-vector space $V_i$ to every vertex $i$ of $Q$ and of a linear operator $A_h : V_i \to V_j$ to every directed edge $h$ going from $i$ to $j$ (loops and multiple edges are allowed). We will show that a representation of a quiver $Q$ is the same thing as a representation of a certain algebra $P_Q$ called the path algebra of $Q$. Thus one may ask: what are the indecomposable finite dimensional representations of $Q$?

More specifically, let us say that $Q$ is **of finite type** if it has finitely many indecomposable representations.

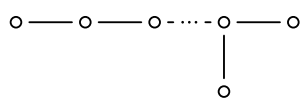
We will prove the following striking theorem, proved by P. Gabriel in early 1970s:[1]

**Theorem 2.1.2.** *The finite type property of $Q$ does not depend on the orientation of edges. The connected graphs that yield quivers of finite type are given by the following list:*

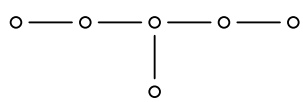
- $A_n$ :



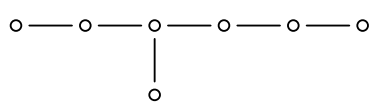
- $D_n$:



- $E_6$



- $E_7$



- $E_8$

[1] We will prove this theorem when the field $k$ is algebraically closed, but it is valid even without this assumption.